MATH F252 Calculus II Syllabus (in person) Fall 2025

Essential Information

		Deven Barnett	Jill Faudree
Instructors		Deven Barnett	Jill Faudree
	section:	902	901
	time:	MWF 11:45am-12:45pm Th 11:30am-1:00 pm	MWF 11:45am-12:45pm Th 11:30am-1:00 pm
	room:	Chap 305	Grue 208
	office:	Chap 303C	Chap 306B
	email:	dcbarnett@alaska.edu	jrfaudree@alaska.edu
	office hours:	MWF 2:00pm-3:00pm Th 1:00pm-3:00pm	MWF 10:30am-11:30am Chap 306B Th 1:00pm-2:00pm (if available) SSC Th 2:00pm-3:00pm SSC

Course webpage https://uaf-math.github.io/calc2/

Canvas site https://canvas.alaska.edu/courses/27071

Prerequisite MATH F251X Calculus I; or placement.

Required text *OpenStax Calculus Volume 2* by G. Strang & E. Herman,
openstax.org/details/books/calculus-volume-2
optional paperback print copy: ISBN-13 978-1-50669-807-6

Description and Course Goals

Calculus is useful, and part of the language, in all the science and engineering disciplines. That is why it is part of the UAF core curriculum. The two principal tools from Calculus I (MATH 251), differentiation and integration, are central here too, but this course extends your understanding of integration especially. We will also study sequences and series, including Taylor series.

At the completion of the course, students will:

- possess mature skills in computing integrals in one variable,
- be able to apply integrals to common problems (areas, volumes, work, . . .),
- understand the convergence of sequences and series,
- understand and be able to apply Taylor series and polynomials, and
- be able to use parametric and polar equations for curves.

In addition, students will have the mathematical foundation for success in Calculus III and other courses requiring knowledge of sophisticated integration techniques in one variable, sequences and series, including many junior/senior-level science and engineering courses.

Class Time

There are 4.5 hours of class meetings with your instructor every week. The 1.5 hour time on Thursday will be used to discuss new topics (30 minutes) and for a weekly Quiz (30 minutes). The final 30 minutes will be spent going over the Quiz. We go over the quiz immediately so that students can leave knowing what quiz material has been mastered, how to work each problem correctly, and what topics need additional work.

Schedule and Online Materials

The course website contains a Schedule listing the textbook sections to be covered each class, the dates each Homework is due, plus the dates for Quizzes and Exams. You should consult this schedule frequently. We will announce any Schedule adjustments in class.

Most course materials (Syllabus, old Quizzes, Exams with solutions, study materials, etc.) will be posted on the course webpage. A few course materials (grades, Homework solutions, announcements, link to Gradescope) will be available on the Canvas site. Each website links to the other.

Office Hours and Communication

Canvas will be used to make announcements for the whole class. Thus, all students should set up their Canvas accounts to receive these announcements in a location that is regularly checked.

Office hours are posted at the top of the syllabus and on the course webpage. You are also welcome to make an appointment.

Evaluation and Grades

Grades are determined as follows. (Each component of the grade is discussed below.)

Homework	15%
Quizzes	20%
Midterm Exam 1	20%
Midterm Exam 2	20%
Final Exam	25%
total	100%

A	93–100%	C	70–72%
A-	90–92%		
B+	87–89%	D+	67–69%
B	83–86%	D	63–66%
B-	80–82%	D-	60–62%
C+	77–79%	F	$\leq$ 59%

The grade ranges at right are a guarantee and a lower bound. We reserve the right to increase your grade above these ranges based on the actual difficulty of the work, average class performance, and/or improvement over the semester include exemplary performance on the final exam.

Homework

Homework assignments consist of a selection of problems from the textbook. All problems for the entire semester can be found at the public webpage here.

The Homework exists because all of us **learn by doing**.

Please write your Homework on paper, or electronically on a tablet, and turned it in as a PDF via Gradescope, accessed via the Canvas page.

Help with scanning homework can be found on the Tech Help webpage. Assignments are due, mostly on Mondays and Wednesdays, by 11:59pm, thus in advance of the Thursday Quiz; see the online Schedule for due dates.

Complete worked solutions to all Homework problems are **provided in advance on the Canvas site**. Thus, Homework will be graded based on effort and completion. However, **the work you turn in is expected to be your own**.

To earn full points on the homework, your submission must:

- be unequivocally your own work
- include work, not bald answers
- be legible, organized, and in order
- be on time

Quizzes

A weekly Quiz will be given on Thursdays. It will be in the middle third of the 1.5 hour, 11:30am-1:00pm class. A Quiz will cover material since the previous Quiz. Quizzes are given under Exam conditions: books, notes, and calculators are not allowed. Performance on Quizzes is your best indicator of how well you are learning the course material, and it is a much better predictor of Exam performance than is your Homework score.

Students will be given the opportunity to correct their Quizzes in the last third of the Thursday class. You can earn-back up to half the missed points for doing so **accurately and thoroughly**.

Always contact your instructor if you will miss a Quiz for a justified reason!

Midterm and Final Exams

There are two Midterms this semester and a cumulative Final Exam:

- **Midterm 1, Thursday October 2**
- **Midterm 2, Thursday November 13**
- **Final Exam, Tuesday December 9, 10:15am-12:15pm**

Midterms are given during the class time. Make-up Midterms will be given only for documented extenuating circumstances, at instructor discretion. Always contact your instructor if you will miss a Midterm, ideally as far in advance as possible.

The location of the Final will be determined near the end of the semester. Department policy does not allow instructors to give an early Final Exam.

All Exams will be closed book, closed notes, and no calculator.

Time Commitment

As with most university courses, you should expect to spend 2-3 hours **outside** of class each week for every credit-hour. So for Calc 2, you should expect to spend a minimum of 8-12 hours actually working homework and examples from the text, asking questions and preparing for quizzes and exams. The 8-12 hours does not include the time spent finding your textbook and homework assignment, getting coffee, etc! Moreover, the study time should be spread throughout the week over a minimum of 5 days.

Getting Help

Calculus 2 is a challenging course which is one of the reasons you will have a sense of accomplishment when you succeed at learning the material. You should expect to experience the discomfort that is the unavoidable part of learning something new. This includes periods of confusion and struggle. You should expect to need to get extra help from your instructor, tutors in the Math and Stat Lab, and each other.

Please come and talk to us if you are unable to clarify points of confusion. We want to know what your long-term struggles are and help you move past them.

Places to Get Help

- The Math and Stat Lab is now located in the brand new student success center on the 6th floor of the Rassmussen Building and offers drop-in tutoring.

  See

  www.uaf.edu/dms/mathlab/ for schedules and availability.
- Free one-on-one (or small group) tutoring is available in Chapman Building Room 201. You must schedule an appointment; see www.uaf.edu/dms/mathlab/.
- Student Support Services (uaf.edu/sss/) offers free tutoring in many subjects to students who qualify for their program.
- ASUAF (uaf.edu/asuaf/) offers private tutoring for a small fee, based on student income.

Rules and Policies

AI usage

Feel free to use a calculator or online tools on Homework. It is also reasonable to explore new AI tools like ChatGPT, but merely doing cut-and-paste without understanding will have no benefit to your learning.

Moreover, during proctored and on-paper Quizzes and Exams, electronic tools of any type are not allowed. Since Quizzes and Exams represent more than 75% of your grade, copying without understanding is not a good long-term strategy.

Incomplete Grade

Incomplete (I) will only be given in DMS courses in cases where the student has completed the majority (normally all but the last three weeks) of a course with a grade of C or better, but for personal reasons beyond his/her control has been unable to complete the course during the regular term. Negligence or indifference are not acceptable reasons for the granting of an incomplete.

Late Withdrawals

A withdrawal after the deadline from a DMS course will normally be granted only in cases where the student is performing satisfactorily (i.e., C or better) in a course, but has exceptional reasons, beyond his/her control, for being unable to complete the course. These exceptional reasons should be detailed in writing to the instructor, department head and dean.

No Early Final Examinations

Final examinations for DMS courses shall not be held earlier than the date and time published in the official term schedule. Normally, a student will not be allowed to take a final exam early. Exceptions can be made by individual instructors, but should only be allowed in exceptional circumstances and in a manner which doesn't endanger the security of the exam.

Academic Dishonesty

Academic dishonesty, including cheating and plagiarism, will not be tolerated. It is a violation of the Student Code of Conduct and will be punished according to UAF procedures.

Student protections and services statement

Every qualified student is welcome in my classroom. As needed, I am happy to work with you, Disability Services, Veterans' Services, Rural Student Services, etc. to find reasonable accommodations. Students at this University are protected against sexual harassment and discrimination (Title IX), and minors have additional protections. As required, if I notice or am informed of certain types of misconduct, then I am required to report it to the appropriate authorities. For more information on your rights as a student and the resources available to you to resolve problems, please go the following site: `www.uaf.edu/handbook/`.

General education statement

This course is listed as a General Education Math Course. As such this course is expected to *contribute to meeting the following* four general learning outcomes:

1. Build knowledge of human institutions, sociocultural processes, and the physical and natural works through the study of mathematics. Competence will be demonstrated for the foundational information in each subject area, its context and significance, and the methods used in advancing each.

2. Develop intellectual and practical skills across the curriculum, including inquiry and analysis, critical and creative thinking, problem solving, written and oral communication, information literacy, technological competence, and collaborative learning. Proficiency will be demonstrated across the curriculum through critical analysis of proffered information, well-reasoned solutions to problems or inferences drawn from evidence, effective written and oral communication, and satisfactory outcomes of group projects.

3. Acquire tools for effective civic engagement in local through global contexts, including ethical reasoning, intercultural competence, and knowledge of Alaska and Alaska issues. Facility will be demonstrated through analyses of issues including dimensions of ethics, human and cultural diversity, conflicts and interdependencies, globalization, and sustainability.

4. Integrate and apply learning, including synthesis and advanced accomplishment across general and specialized studies, adapting them to new settings, questions and responsibilities,

and forming a foundation for lifelong learning. Preparation will be demonstrated though production of a a creative or scholarly product that requires broad knowledge, appropriate technical proficiency, information collection, synthesis, interpretation, presentation and reflection.

Official UAF Syllabus Addendum

Student protections statement: UAF embraces and grows a culture of respect, diversity, inclusion, and caring. Students at this university are protected against sexual harassment and discrimination (Title IX). Faculty members are designated as responsible employees which means they are required to report sexual misconduct. Graduate teaching assistants do not share the same reporting obligations. For more information on your rights as a student and the resources available to you to resolve problems, please go to the following site:
https://catalog.uaf.edu/academics-regulations/students-rights-responsibilities/.

Disability services statement: I will work with the Office of Disability Services to provide reasonable accommodation to students with disabilities.

Student Academic Support:

- Speaking Center (907-474-5470, uaf-speakingcenter@alaska.edu, Gruening 507)
- Writing Center (907-474-5314, uaf-writing-center@alaska.edu, Gruening 8th floor)
- UAF Math Services, uafmathstatlab@gmail.com, Chapman Building (for math fee paying students only)
- Developmental Math Lab, Gruening 406
- The Debbie Moses Learning Center at CTC (907-455-2860, 604 Barnette St, Room 120, https://www.ctc.uaf.edu/student-services/student-success-center/)
- For more information and resources, please see the Academic Advising Resource List (https://www.uaf.edu/advising/lr/SKM_364e19011717281.pdf)

Student Resources:

- Disability Services (907-474-5655, uaf-disability-services@alaska.edu, Whitaker 208)
- Student Health & Counseling [6 free counseling sessions] (907-474-7043, https://www.uaf.edu/chc/appointments.php, Whitaker 203)
- Center for Student Rights and Responsibilities
(907-474-7317, uaf-studentrights@alaska.edu, Eielson 110)
- Associated Students of the University of Alaska Fairbanks (ASUAF) or ASUAF Student Government (907-474-7355, asuaf.office@alaska.eduasuaf.office@alaska.edu, Wood Center 119)

ASUAF advocacy statement The Associated Students of the University of Alaska Fairbanks, the student government of UAF, offers advocacy services to students who feel they are facing issues with staff, faculty, and/or other students specifically if these issues are hindering the ability of the student to succeed in their academics or go about their lives at the university. Students who wish to utilize these services can contact the Student Advocacy Director by visiting the ASUAF office or emailing asuaf.office@alaska.edu.

Nondiscrimination statement: The University of Alaska is an affirmative action/equal opportunity employer and educational institution. The University of Alaska does not discriminate on the basis of race, religion, color, national origin, citizenship, age, sex, physical or mental disability, status as a protected veteran, marital status, changes in marital status, pregnancy, childbirth or related medical conditions, parenthood, sexual orientation, gender identity, political affiliation or belief, genetic information, or other legally protected status. The University's commitment to nondiscrimination, including against sex discrimination, applies to students, employees, and applicants for admission and employment. Contact information, applicable laws, and complaint procedures are included on UA's statement of nondiscrimination available at www.alaska.edu/nondiscrimination. For more information, contact:

UAF Department of Equity and Compliance

1760 Tanana Loop, 355 Duckering Building, Fairbanks, AK 99775

907-474-7300

uaf-deo@alaska.edu

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